1. Indicate which Mathematical Theorem *best fits* the statement.

    Fermat's Theorem
    Intermediate Value Theorem
    Extreme Value Theorem
    Mean Value Theorem

    - There was a maximum temperature in the last 24 hours.
    - A ball tossed in the air will be at its highest point when it stops moving for an instant.
    - A fish that jumped above the surface of the water will make a splash before it is again underwater.
    - If you travel 90 miles in 2 hours, at some time you were traveling 45 miles per hour.

2. Recall that Fermat's Theorem states
    "If *f* has a local minimum or maximum value at *c*, and if $f'(c)$ exists, then $f'(c)=0$."

    TRUE OR FALSE: It is also true that this statement can be reversed. That is to say
    "If $f'(c)=0$, then f has a local minimum or maximum value at *c*."

3. A person begins cycling from home at noon. She begins cycling quickly and travels 3 miles north. Then she makes a U-turn without stopping, and then cycles home slowly. She finishes cycling after 30 minutes. Let $g(x)$ be a function that represents the total displacement of the biker from her home versus time.

a) Construct and label a possible graph of $g(x)$.
b) Construct a line segment on the graph from part (a) from the starting point to the ending point. On the graph, label the point *c* guaranteed by the appropriate theorem.
c) Refer to the cycling context of the problem to describe what the point *c* means in the problem.

4. Let $g(x)=x^2-4x+3$.

a) If $a=0$ and $b=2$, what value *c* satisfies the conclusion of the MVT?
b) If $a=0$ and $b=3$, what value *c* satisfies the conclusion of the MVT?
c) If $a=1$ and $b=3$, what value *c* satisfies the conclusion of the MVT?
d) In general, what value *c* do you anticipate satisfies the conclusion of the MVT for any *a* and *b* with this polynomial?
e) Suppose that $p(x)=Ax^2+Bx+C$. Show that the number *c* that satisfies the conclusion of the MVT for any polynomial $p(x)$ is given by $c=\dfrac{a+b}{2}$

5. Use Rolle's Theorem & the Intermediate Value Theorem to show that the equation $6x - \cos(x) = 0$ has exactly one real solution.

6. Suppose *f* is an odd function and is differentiable everywhere. Prove that for every positive number *b*, there exists a number *c* in $(-b, b)$ such that $f'(c) = \frac{f(b)}{b}$.

7. Suppose that *f* and *g* are continuous on $[a, b]$ and differentiable on (a, b). Suppose also that $f(a) = g(a)$ and $f'(x) < g'(x)$ for $a < x < b$. Prove that $f(b) < g(b)$.